

8-14-2015

A New SCADA Dataset for Intrusion Detection System Research

Ian P. Turnipseed

Follow this and additional works at: <https://scholarsjunction.msstate.edu/td>

Recommended Citation

Turnipseed, Ian P., "A New SCADA Dataset for Intrusion Detection System Research" (2015). *Theses and Dissertations*. 209.

<https://scholarsjunction.msstate.edu/td/209>

This Graduate Thesis - Open Access is brought to you for free and open access by the Theses and Dissertations at Scholars Junction. It has been accepted for inclusion in Theses and Dissertations by an authorized administrator of Scholars Junction. For more information, please contact sct@library.msstate.edu.

A New SCADA Dataset for Intrusion Detection System Research

A new SCADA dataset for intrusion detection system research

By

Ian Turnipseed

A Thesis
Submitted to the Faculty of
Mississippi State University
in Partial Fulfillment of the Requirements
for the Degree of Master of Science
in Electrical and Computer Engineering
in the Department of Electrical and Computer Engineering

Mississippi State, Mississippi

August 2015

Copyright

Ian Turnipseed

2015

A new SCADA dataset for intrusion detection system research

By

Ian Turnipseed

Approved:

Thomas H. Morris
(Major Professor)

David A. Dampier
(Committee Member)

Robert Wesley McGrew
(Committee Member)

James E. Fowler
(Graduate Coordinator)

Jason M. Keith
Dean
Bagley College of Engineering

Name: Ian Turnipseed

Date of Degree: August 14, 2015

Institution: Mississippi State University

Major Field: Electrical and Computer Engineering

Major Professor: Dr. Thomas Morris

Title of Study: A new SCADA dataset for intrusion detection system research

Pages in Study: 57

Candidate for Degree of Master of Science

Supervisory Control and Data Acquisition (SCADA) systems monitor and control industrial control systems in many industrials and economic sectors which are considered critical infrastructure. In the past, most SCADA systems were isolated from all other networks, but recently connections to corporate enterprise networks and the Internet have increased. Security concerns have risen from this new found connectivity. This thesis makes one primary contribution to researchers and industry. Two datasets have been introduced to support intrusion detection system research for SCADA systems. The datasets include network traffic captured on a gas pipeline SCADA system in Mississippi State University's SCADA lab. IDS researchers lack a common framework to train and test proposed algorithms. This leads to an inability to properly compare IDS presented in literature and limits research progress. The datasets created for this thesis are available to be used to aid researchers in assessing the performance of SCADA IDS systems.

DEDICATION

Thanks for putting me through college Mom and Dad. I would also like to thank Caterina for her continuous support along the way ☺. Lastly, I would like to mention all the friends and professors I have met and my brothers and sister, thanks for your guidance!

ACKNOWLEDGEMENTS

I would like to first acknowledge Dr. Morris for all his support and knowledge that he gave me in my travels through graduate school. I would like thank him also for his guidance into the research area of ICS security. It has helped me tremendously and will never be forgotten. I would like to also thank both of my committee members Dr. Dampier and Dr. McGrew for both the knowledge I have gained from them as professors and for their support in graduate school.

TABLE OF CONTENTS

DEDICATION	ii
ACKNOWLEDGEMENTS	iii
LIST OF TABLES	vi
LIST OF FIGURES	vii
CHAPTER	
I. INTRODUCTION	1
1.1 Background.....	1
1.2 Research Contributions	5
1.3 Organization	6
II. LITERATURE REVIEW	7
2.1 SCADA System Threats.....	7
2.2 Intrusion Detection	10
2.3 SCADA Datasets and Test Beds	15
III. GAS PIPELINE DATASET.....	17
3.1 Introduction	17
3.2 Previous Work	18
3.3 Gas Pipeline System	19
3.4 Dataset Collection Methodology	22
3.5 Dataset Description	27
3.5.1 Raw Dataset.....	28
3.5.2 ARFF Dataset	30
3.6 Dataset Validation	41
IV. CONCLUSIONS	52

REFERENCES	53
APPENDIX	
A. REGISTER MAPPING FOR GAS PIPELINE RTU	56

LIST OF TABLES

2.1	Common SCADA System Threats[10]	8
3.1	Attack Categorization.....	24
3.2	Feature List.....	31
3.3	System mode features.....	33
3.4	Feature list	35
3.5	Categories of attacks [7].....	36
3.6	Cyber attacks 1-12.....	37
3.7	Cyber attacks 13-23.....	39
3.8	Cyber attacks 24-35.....	40
3.9	List of Algorithms	41
3.10	Results of Algorithms.....	42
3.11	Percentage of attacks in dataset.....	43
3.12	Comparison of False Positive Rates	44
3.13	Precision and Recall for Datasets	45
3.14	Confusion Matrix for NMRI and CMRI attacks	46
3.15	Identified Problems in the Feature list of Gao's Dataset [26]	50

LIST OF FIGURES

1.1	Simple SCADA system [29]	2
2.1	SCADA Attack Patterns [1]	9
3.1	Gas Pipeline System and HMI	20
3.2	Modbus RTU and TCP packet [21]	21
3.3	Types of Cyber Threats[27].....	23
3.4	Process Framework/Methodology.....	26
3.5	Modbus Frame	28
3.6	Instance within Raw Dataset	30
3.7	Differences between Setpoint Values.....	48
3.8	Gain Parameter Coverage.....	49
3.9	Differences between Control Modes	50
A.1	Register Mapping Sheet	57

LIST OF ABBREVIATIONS

ADU	– Application Data Unit
ARFF	– Attribute Relationship File Format
BSM	– Basic Security Mode
CIP	– Critical Infrastructure Protection
CMRI	– Complex Malicious Response Injection
CRC	– Cyclic Redundancy Code
CSV	– Comma Separated Value
DARPA	– Defense Advanced Research Projects Agency
DNP3	– Distributed Network Protocol
DoS	– Denial of Service
FC	– Function Code
FIFO	– First In First Out
GUI	– Graphical User Interface
HIDS	– Host Intrusion Detections System
HMI	– Human Machine Interface
HVAC	– Heating, Ventilation, and Air Conditioning
ICS	– Industrial Control System
IDS	– Intrusion Detection System
IP	– Internet Protocol

IT – Information Technology

LRC – Linear Redundancy Code

MBAP – Modbus Application Header

MFCI – Malicious Function Code Injection

MLA – Machine Learning Algorithm

MPCI – Malicious Parameter Command Injection

MSCI – Malicious State Command Injection

MTU – Master Terminal Unit

NIDS – Network Intrusion Detection System

NIPS – Network Intrusion Protection System

NMRI – Naïve Malicious Response Injection

NNIDS – Network Node Intrusion Detection System

PART – Projective Adaptive Resonance Theory

PDU – Protocol Data Unit

PID – Proportional-Integral-Derivative

PLC – Programmable Logic Controller

PSI – Pounds per Square Inch

R2L – Remote to Local

RTU – Remote Terminal Unit

SCADA – Supervisory Control and Data Acquisition

TCP – Transmission Control Protocol

U2R – User to Remote

WEKA - Waikato Environment for Knowledge Analysis

CHAPTER I

INTRODUCTION

1.1 Background

Supervisory Control and Data Acquisition (SCADA) systems monitor and control highly critical infrastructure related utilities. These SCADA controlled systems are gas pipelines, power plants, railroads, water treatment facilities, and even some HVAC systems. In the past, most of these systems were isolated from all other networks, but recently they many been integrated with the Internet and corporate enterprise networks. By interconnecting these systems with other networks, control has increased for the operators, and savings have increased for companies. With this newfound connectivity, however, there are also many security concerns for these once isolated and remote systems. If a vulnerability exists in one of these systems, it will now allow attackers to remotely exploit and take control of these SCADA systems; this could cause failure in the hardware and harm to people's lives.

SCADA systems provide control and visualization of critical infrastructure systems. These systems are generally made up of four components [1]. The first level consists of sensors and actuators [1]. The sensors which collect data about the system are pressure monitors, water level gauges, and laser sensors. The actuators control the system's state: pumps, motors, etc. The second level is the programmable logic controllers (PLCs). These components control and collect information that determine the

system's state. The controllers are generally referred to as remote terminal units (RTUs). The RTU interfaces with the first level of the SCADA system, i.e. stores the sensor data in predetermined registers. The third level of a SCADA system is supervisory controls [1]. The supervisory controls are usually handled by the master terminal unit (MTU). The MTU is the unit which communicates with the RTU. For example, in a gas pump system the MTU can send a command to the RTU to turn on the pump. The MTU can also send a read query to read from the RTU registers that contain the current pressure measurement. There are many protocols which allow for this communication such as Profibus, Fieldbus, Modbus, and Distributed Network Protocol Version 3 (DNP3). The fourth level, the human-machine interface (HMI), is used by an operator to display the sensor data collected by the MTU. The HMI usually contains a visual representation of the system and the operation of the subsystems. It is also used to change parameters and states within the SCADA system with communication through the MTU. A simple SCADA system is shown in the diagram below.

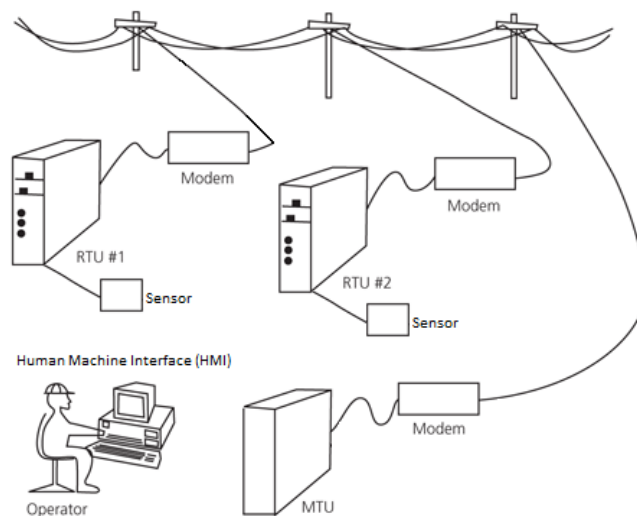


Figure 1.1 Simple SCADA system [29]

According to "Security Aspects of SCADA and Corporate Network Interconnection: An Overview", SCADA systems “were designed to be open, robust, and easily operated and repaired, but not necessarily secure” [1]. This architecture suffers from three key security factors. The first factor is the lack of authentication of the communication protocols used by these SCADA systems [1]. This can lead to spoofing of data transmitted by both the MTU and the RTU. This type of attack occurs at the protocol level and is hard for an Intrusion Detection System (IDS) to detect. The second factor is the idea that these systems are “secure through obscurity” [1]. This means that the operators of these systems believe their specialized equipment and protocols cannot be understood by anyone outside of their knowledge group [1]. The last factor is the notion that the system cannot be harmed by an intruder because it is physically secure [1]. These flawed ideas have made critical infrastructure systems vulnerable and in need of improved cyber security protections.

There are many researchers examining the security in today’s SCADA systems to help bring to light possible vulnerabilities and provide security solutions. A recent attack in 2010, named Stuxnet, was able to compromise uranium-enrichment plants in Iran by targeting the Siemens Step7 software [2]. The Siemen’s software was used to program PLCs, the digital devices that control the industrial systems. Stuxnet was introduced into the Windows environment and began searching for the Siemen’s software [2]. According to “How Stuxnet Is Rewriting the Cyberterrorism Playbook,” once the software was detected, Stuxnet was able to collect surveillance data, place the system into a critical state, and even falsely respond to prevent alarms [2]. It was able to accomplish this by

overwriting the ladder logic and firmware on the PLC, allowing for the attacker to force the PLC to report false responses [3].

Another attack known to have targeted SCADA systems is Flame which collected surveillance data. Flame is similar to Stuxnet in that it infects Windows based systems. The difference is that Flame does not want to cause harm to the system, but rather it gathers and streams data to the command and control server [4]. The server would then filter through the large amount of data being sent from the victim's system and display the results to an operator [4]. This attack was mostly exploited on systems in Iran to collect data for some unknown nation state [4].

Another recent event, Aurora, was presented to the government by Idaho National Laboratory as a demonstration to express the seriousness of these types of attacks. The Aurora exploit was used on an experimental platform which replicated the controls of a power system [5]. The attack specifically targeted the control system of a power system and attempted to open and close circuit breakers [5]. The end goal was to cause a generator to be physically damaged due to a change in the operation cycle causing a fatal phase condition [5]. Although this attack was not exploited in a real system, the goal of grabbing the attention of the government was accomplished, and research and development in industrial control system (ICS) security has increased.

With the use of an intrusion detection systems (IDS), attacks such as these can be detected, and an operator can be alerted to anomalous activity to help prevent further damage. IDSs are an important layer of security which can be implemented in any communication based system to monitor and analyze a system's status. IDSs in SCADA systems are improved by training them with data logs that represent real SCADA

network traffic. The need for a dataset which can be used to validate and improve IDS systems is in high demand. The next sections will describe the research contributions and provide an overview of this thesis.

1.2 Research Contributions

This thesis makes one primary contribution to researchers and industry. The contribution is that of two datasets which have been introduced to replace a previous dataset [6], hence the Gao dataset, that was deemed unsuitable for IDS research. The information gathered in the dataset is network transactions between a Remote Terminal Unit (RTU) and a Master Control Unit (MTU) in Mississippi State University's in-house SCADA gas pipeline. The new datasets were collected using a novel framework for simulating real attacks and operator activity on a gas pipeline. Comparison to the previous dataset validated that all problems which effected that dataset have been resolved.

The datasets contain three separate categories of features: network information, payload information, and labels. The network information provides a pattern of communication for intrusion detection systems to train against. Unlike Information Technology (IT) networks, SCADA systems have network topologies which are fixed and the transactions between the nodes are repetitive and regular. This static behavior is conducive to IDSs to detect anomalous activity. The second category of features is that of payload information. The payload information provides information about the gas pipeline's state, settings, and parameters. These values are vital to understanding how the system is performing and detecting if the system is in an out-of-bounds or critical state.

The last category of features is the network transaction label. The label is appended to each line in the dataset to detail if the transaction is normal system activity or an attack.

It is postulated that the datasets can be used to aid researchers in assessing the performance of SCADA IDS systems through the use of real SCADA attack patterns and simulated operator operation of the HMI. Since SCADA systems are designed to last several years [7], the network topologies are predetermined and communication patterns remain fixed. Therefore these datasets can be used to assess SCADA IDS systems in general, by providing these common characteristics.

1.3 Organization

The remainder of this thesis is arranged as follows. The next chapter, provides recent research in the areas of SCADA system threats, IDSs for critical infrastructure systems, and an analysis of other SCADA datasets and test beds. This chapter analyzes exactly why this dataset is needed in the research community and what purpose it will serve to other researchers. Chapter III discusses the gas pipeline system that was used to create this dataset along with the methodologies and framework that were implemented. The next two sections in Chapter III detail the two datasets which were created. The first is the raw network transaction data which was captured, while the second preprocesses the data given from the first. There is also a section which validates that the current dataset has improved from the previous iteration. The last chapter provides conclusions which were made from this research.

CHAPTER II

LITERATURE REVIEW

2.1 SCADA System Threats

Many researchers are studying and creating IDS systems using SCADA network traffic. SCADA systems are becoming increasingly vulnerable to outside attackers and have plagued the minds of security professionals. In “Security for Process Control, An Overview,” Brundle and Naedele [8] discuss the importance of securing industrial control systems by listing the challenges of SCADA security, providing reactions from the industry to these challenges, and highlighting what security threats remain. There are several security threats which remain in SCADA systems as shown in [8][9][10]. In “Challenges and Direction toward Secure Communication in the SCADA System,” Hong and Lee discuss the inherent security issues in SCADA and smart grid communication technologies [9]. They provide details on how these open standard communication protocols are becoming more vulnerable to cyber-attacks as they are connected to larger networks. Since these protocols were made to be on isolated networks there are security vulnerabilities not addressed in these protocols which do not account for being connected to larger networks. Hong and Lee also discuss some of the problems with intrusion detection systems. They state that SCADA IDSs require network traffic patterns to determine if there is any anomalous activity occurring within the system [9]. The need for a dataset which represents a real SCADA system and contains the “peculiar” traffic is

in high need to create IDSs tailored for SCADA applications. In “Analysis on Cyber Threats to SCADA Systems” by Kang et al. also discusses many problems in current SCADA systems [10]. The table below details the many attacks that effect these systems.

Table 2.1 Common SCADA System Threats[10]

Common RT Computer System Threats			
1. Authorization Violation	9. Information Leakage	17. Sabotage	25. Traffic Analysis
2. Bombs (Logic or Time)	10. Intercept/Alter	18. Scavenging	26. Trap Door/ Back Door
3. Browsing	11. Interference Database Query Analysis	19. Spying	27. Trojan Horse
4. Bypassing Controls	12. Masquerade	20. Service Spoofing	28. Tunneling
5. Data Modification	13. Physical Intrusion	21. Sniffers	29. Unauthorized Access Violations of Permission
6. Denial of Service	14. Replay	22. Substitution	30. Unauthorized Access
7. Eavesdropping	15. Repudiation	23. Terrorism	31. Virus
8. Illegitimate Use	16. Resource Exhaustion	24. Theft	32. Worm

Many of these attacks are included in the proposed dataset and can be used to effectively train IDSs to detect attacks similar to these.

There are many other reports which detail security vulnerabilities in SCADA systems [11][12]. In a report published by Dell on SCADA attack patterns, the authors identified the most common threats common to SCADA systems [11]. The diagram below shows their results.

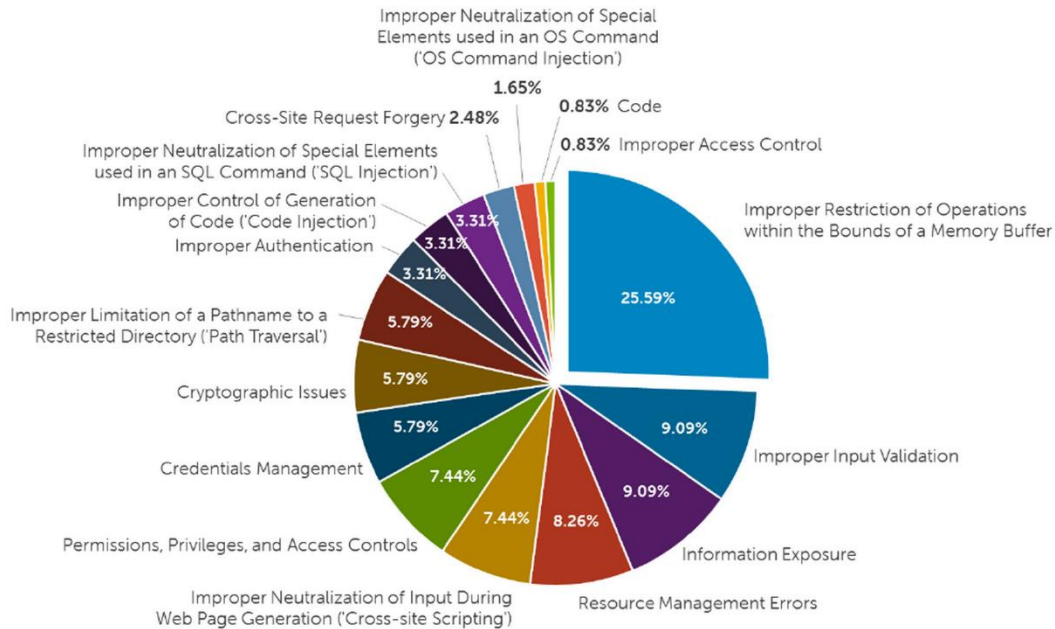


Figure 2.1 SCADA Attack Patterns [1]

These attacks are executed to gain access to servers which control the SCADA systems. Once these servers are compromised, an attacker has access to the workstations which operate the physical process. “Software security: Application-level vulnerabilities in SCADA systems” by Valentine et al. addresses what can happen when these workstations are compromised. They discuss how ladder logic used in the PLCs of SCADA systems do not provide protection against many common errors [10]. They provide many examples of intentional and unintentional errors at the application level. Their conclusions showed that verification and validation tools should be added to provide another layer of protection for the PLCs. Similar to Hong and Lee’s work, Dzung et al. outlines in “Security for Industrial Communication Systems” the many issues found in communication networks for industrial applications [13]. Dzung provides

a list of attacks that are common to this application domain. These attacks are a subset of the ones found in Table 2.1. In conclusion Dzung states that it is possible to secure industrial control systems using many emerging and conventional technologies. One of Dzung's security technologies that is recommended is intrusion detection systems. These systems are crucial to providing real time information of anomalous or malicious activity. The next section will discuss intrusion detection systems.

2.2 Intrusion Detection

Intrusion detection systems are used to collect and analyze system activity data to monitor a system's status. They also provide examination of a system's state and perform integrity checks on files within the system. Many IDSs use machine learning algorithms for pattern recognition to detect threat activity which is anomalous for a certain system. There are other IDSs which use a signature-based system to compare activity to a database of known threats [14]. These functionalities can be combined together for a robust detection system and will provide a sufficient layer of protection for various attacks.

An IDS consists of three main components. The first component is the "Network Intrusion Detection system (NIDS)" [14]. The NIDS uses a signature-based system to determine if the activity in the entire system is normal or if it can be found in the database of known attacks [14]. If the NIDS finds a match of a signature, the activity is reported to the operator or system administrator [14]. The NIDS does not prevent the traffic from going through, but only provides a warning. The second component is the "Network Node Intrusion detection system (NNIDS)" [14]. The NNIDS is more specific to the communication between the control station and a single subsystem. It performs similar

functions as the NIDS, but it also provides some pattern recognition and behavior analysis. Each subsystem requires an algorithm that is tailored toward its functions; this is the reason for pattern recognition to be on this level. The specialization is needed to provide the highest level of security and detection. The last level of security is found on the subsystem itself. It is referred to as the “Host Intrusion Detection System (HIDS)” [14]. This is the system that analyzes system state and performs integrity checks on the systems data to determine if there is anomalous activity. Many SCADA systems have low variability in states, and changes from normal behavior are easily detected with a proper IDS installed.

There are many limitations to the IDS solutions discussed above, such as a high false positive rate from noise generated in normal activity. The noise is introduced from the system in the form of a bad packet or a hardware malfunction, but it would be detected as anomalous and reported to the operator as a possible threat [16]. This high number of false positives can reduce the effectiveness of the IDS, as the false positives will overshadow the actual threats often causing real warnings to be ignored [16]. Another problem with IDS solutions, specifically the signature-based solutions, is the constant need to upgrade signatures [16]. Although most systems require updating, this can sometimes be overlooked by operators and can cause the systems to be vulnerable to the most recent exploits. The next limitation is that the IDS cannot secure a system that has poor authentication and identification protocols [16]. This is because of spoofing, the introduction of normal behavior into the system that is identical to the traffic of the real control station. For example spoofing, can be a problem with a system which records pressure data, as it can allow false reports of the actual pressure measurement and cause

the operator to correct a system that is not actually in need of a control action, possibly over-pressurizing the system. The above example is similar to the Aurora attack that was designed for a power system and caused a generator to be in a critical condition. The attack was able to use legitimate requests to place the generator into an out-of-phase state that would cause harm to the system [5]. The last limitation that will be discussed is the problem with analyzing encrypted traffic. An IDS is not able to inspect packets deeply if the traffic is encrypted; therefore, traffic must be unencrypted before any attempts to analyze it occur. The problem lies in processing times and may inhibit the IDS from performing in real time. Although there are some limitations of an IDS, it still performs an important role in securing networks.

IDSs are commonly used in computer networks (firewalls, etc.) and virus software, but they are emerging in the industrial control system world. These systems have been important in protecting web servers and personal computers and are now a field of research for ICS professionals. Many reasons to implement an IDS in SCADA systems have been discussed above. For example, highly critical infrastructure is dependent on many specialized protocols that are designed with ease of use and reparability in mind, rather than security [15]. These systems are highly reliant on the operators and are in need of an automated approach to monitoring the system's normal activity. There are many studies to provide a broad approach which can be easily distributed to system operators to improve security.

A recent approach for an IDS to be implemented on SCADA systems was demonstrated in “An unsupervised anomaly-based detections approach for integrity attacks on SCADA systems”, by Abdulmohsen Almalawi et. al [17]. Almalawi proposed

that an unsupervised learning algorithm (pattern recognition) approach would perform best on SCADA network traffic [17]. He tested the theory by using data from a real world industrial system, a water plant, and experimented with two different types of algorithms [17]. Almalawi used many pre-processing techniques, massaging the input to improve results in an algorithm while preserving the integrity of the data, to control noise that was within the dataset from the water plant [17]. Almalawi then processed the dataset through his clustering algorithm, a behavior analysis technique [17]. He concluded that his type of behavioral approach shows promise and can accomplish high detection results within this field [17]. With the fixed-width algorithm, Almalawi was able to achieve a detection rate of over 90% with less than .01% false-negatives. The only concern was the processing time, and complexity of the algorithms, and the data set contained no examples of attack behavior and should be researched further to improve the efficiency of the process [17].

Another example of this automated approach is outlined in “Improving Security for SCADA Sensor Networks with Reputation Systems and Self-Organizing Maps”, by Jose M. Moya et. al [18]. Moya also used unsupervised learning algorithms, but he took into account the severe processing power that is required to use these techniques. Moya first trained the algorithm with a dataset of only normal activity [18]. This built clusters, which he can then use to classify anything outside of these normal activity clusters as anomalous. By creating the clusters before operation, Moya reduced the processing that is required during normal operation of the IDS. He combined this with a quantization error which can decipher attack traffic that is similar to normal traffic [18]. The quantization error is calculated from the distance of that specific packet to the centroid of the cluster [18]. If the distance is beyond a threshold, it is considered anomalous. Attack traffic

similar to normal is a common problem when spoofing is involved, as it can create legitimate requests that are identical in structure to normal traffic. The quantization error is able to place many spoofed packets in the anomalous clusters rather than cause a false-negative [18].

There are some products which are already used in many real life systems and do not use the novelty approach above. Many of these solutions require signature databases or rules generated by the operator. The Snort IDS is an example of one of these products. Snort IDS is a NIDS that is able to log and analyze real time SCADA network traffic [19]. Snort has the capability of examining network packets and also deep packet inspection, the ability to explore information within the payload of a packet. This type of product is dependent upon a ruleset defined by the operator of a system [19]. The ruleset is either a database of signatures or rules created by a professional in the field [19]. “Snort is, by far, the gold standard among open source NIDS systems, with over 100,000 users and 3 million downloads to date” [19]. The Snort IDS is also free, which allows any company to install this type of system to increase the number of security layers within their systems [19]. This type of IDS is effective for known attacks, but struggles with attacks that are similar to normal traffic and places the system in states that are defined.

Another product that is used for IDS is Bro. Bro is not commonly used in commercial systems, but rather for research. Bro can be adjusted to work with almost any computer based communication protocol. In “Adapting Bro into SCADA: Building a Specification-based Intrusion Detection System for the DNP3 Protocol”, by Hui Lin et al. is able to use Bro in a SCADA system that uses the DNP3 protocol [20]. DNP3 is a commonly used communication protocol in SCADA type systems. Bro uses a similar

detection scheme as Snort, but instead of using a ruleset, it uses known signatures of attacks. Lin uses Bro to recognize semantics and provide validation of the DNP3 protocol in the SCADA traffic [20]. Bro provides detection on attacks that replay previous packets for denial of service [20]. Protection is also provided for created cyclic redundancy check errors and attacks that cause unstable system states [20]. The denial of service attacks can be detected just from observing the patterns that these attacks exhibit [20]. Recognition of unstable state attacks requires knowledge about the system. Bro and Snort both require databases for signatures. Almalawi and Moya both use machine learning algorithms which do not require a database to understand the difference between normal and anomalous behavior. Their approaches train MLAs against a dataset which automatically categorizes or clusters the data into these categories. Both of these approaches use completely independent datasets to test their methods which makes comparison of the two approaches difficult. The next section will discuss many of the datasets and test beds that are available for researchers to use and why the proposed dataset is needed.

2.3 SCADA Datasets and Test Beds

SCADA datasets and test beds can be used to analyze the performance of IDSs. IDS researchers lack a common framework to train and test proposed algorithms. This leads to an inability to properly compare proposed IDSs and limits research progress. Many of the datasets used by researchers do not contain all types of attacks and gauging the performance of the IDS is hard when all patterns of attack are not considered. In [17][18], Almalawi and Moya use separate datasets to test the performance of their IDS. In the case of Almalawi, the dataset used in his research was from a water treatment plant. Although his dataset was from a real world system, it was not possible for him to

run attacks or simulate a fault against a system which is providing services to the world. These unknowns make it difficult to gauge the effectiveness of his IDS. Similarly, Moya did not provide an in depth discussion of the data used in his research. He states the data is from a simulated sensor network and it contains attack patterns within. There are several other researchers providing IDSs to the community by using their individual dataset [22][23][24]. In “Building a SCADA Security Testbed” by Mahmood et al describes his testbed which is meant to provide a simulation of a real SCADA system and is able to connect to multiple real world systems [22]. This test bed allows researchers to run attacks against his model and test IDSs against attacks. The problem with the test bed is that it is not able to provide a dataset that is from a real system and the attacks run against the system may not be comprehensive. In [23] “Using Model-based Intrusion Detection for SCADA Networks” by Cheung et al, the researchers used a dataset collected on a SCADA test bed located at Sandia National Laboratories. This dataset is said to only contain reconnaissance type attacks on the Modbus TCP protocol. There IDS is not being tested against other categories of attacks including injection and denial of service (DoS). In [24] “Anomaly-Based Intrusion Detection for SCADA Systems” Yang et al use a dataset which was collected on an Idaho National Lab’s simulation of a SCADA system. The dataset contains both DoS and injection type attacks, but does not include the reconnaissance attacks. A common dataset is needed to provide third-party validation of IDS solutions. The dataset from this research has been created to fill the void in this area. The next chapter will provide a detailed description of the dataset created.

CHAPTER III

GAS PIPELINE DATASET

3.1 Introduction

The 1999 DARPA dataset produced by MIT's Lincoln Labs [25] was created with the intent for researchers to test viable Intrusion Detection Systems (IDS) for effectiveness. The dataset has been a vital part in furthering research for evaluating computer network IDSs and provides a benchmark for other researchers to compare and validate results. The dataset was collected from a simulated Air Force base network connected to the Internet. The simulated network produced a dataset that contained network traffic in the form of a tcpdump. The information within the dataset includes the sniffed network packets, Sun BSM data, file system information, and process information for the purpose of identifying anomalous behavior contained within the collected timeframe.

The various attacks which were carried out on the system are thoroughly explained in the thesis of Kendal [25], but a brief explanation will be provided below. The attacks are categorized into five groups: Data Attacks, User to Remote (U2R), Remote to Local (R2L), Denial of Service (DoS), and Probe. The data attacks were used to extract files in which the security policy states that the files should remain on the host computer. This means that files that were secret or confidential were allowed to leave the computer which was accessed by a legitimate user. The U2R attacks allowed a local user

to elevate privileges to communicate to a remote location. The R2L attacks allowed an attacker to gain access to a victim's machine and extract files or modify data in transit from the victim's machine. DoS attacks were designed to disrupt transmission of data from network services. The last category of attacks included in the DARPA dataset are probing attacks. This category of attacks aimed to collect information about the specific system, such as IP addresses of local machines, open ports, and local operating systems. By including attacks and normal activity within the dataset, researchers are able to create IDSs tailored towards these applications. The same holds true for SCADA systems, but currently a dataset does not exist which is accessible to all researchers and includes various types of attacks.

3.2 Previous Work

The dataset proposed and created for this research is a second iteration of a previous dataset from a gas pipeline system to fill the void in IDS research for SCADA applications. The first iteration of the dataset was created by Wei Gao [7]. Gao's dataset was found to contain obvious patterns, which caused algorithms to appear to have extremely high detection rates, up to 100%. The paper by Thornton et al. was initially written to determine if machine learning algorithms could be used for anomaly detection in SCADA systems. They also wanted to determine how effective these machine learning algorithm by testing them with the Gao dataset. but instead, found the datasets contained many serious issues [26]. The conclusions of the paper showed that the dataset was unsuitable for IDS research due to obvious correlations between particular parameters and the result to be predicted by the algorithms. These correlations are

unrealistic in real SCADA transactions and renders the datasets unsuitable in their current form [26].

Many of these unrealistic transactions were caused by the system being placed into only three different state configurations. To rid the dataset of these obvious patterns, a new process was created to place the system into all possible state configurations that represent normal operation of the gas pipeline. The states were chosen in random order to reduce the chance of unintended patterns. Another factor that caused obvious patterns within the dataset was the invariable attacks that were run against the system. The attacks were static and did not contain dynamically changing parameters. The new process of collecting the dataset also addressed this problem by parameterizing and randomizing the order in which the attacks were executed. New attacks were also created in conjunction with the existing attacks created by Gao [7]. The gas pipeline system which was used in the creation of these datasets is discussed below.

3.3 Gas Pipeline System

The gas pipeline system used to collect the datasets was provided by Mississippi State University's in-house SCADA lab. The system consists of three major components: sensors and actuators, a communication network, and supervisory control. Below is a picture of the physical system along with the iFIX HMI.

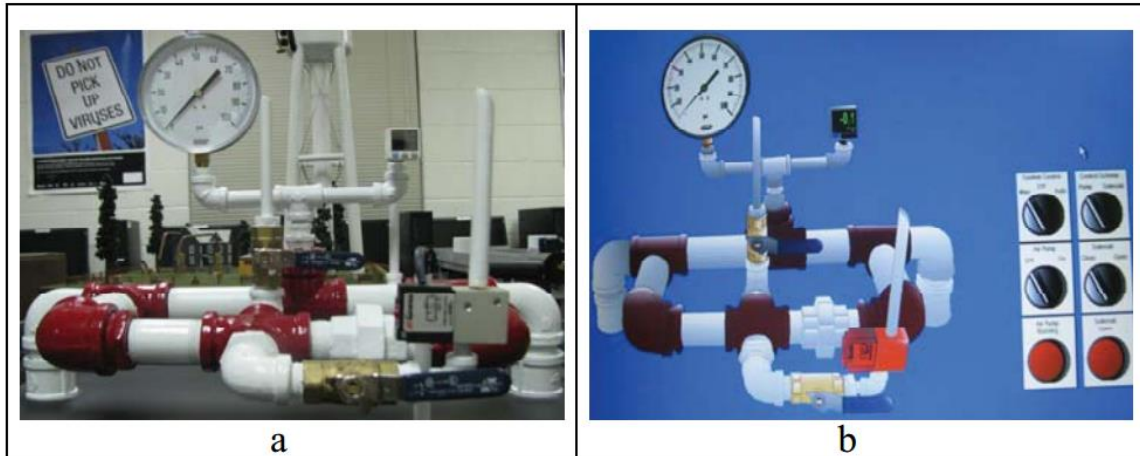


Figure 3.1 Gas Pipeline System and HMI

At the lowest level, the gas pipeline contains two actuators along with a pressure sensor. The actuators, a pump and a solenoid, are used to control the physical process of the system, to maintain the pressure set by the supervisory controls. The gas pipeline has three main system modes: automatic, manual, off. When the system is in automatic mode, there are two schemes to maintain the pressure which are decided by the supervisory controls. The first scheme is pump mode, which turns the pump on and off to keep the pressure in the pipe at the set point. This scheme was created to simulate a constant load on the system. The second scheme is solenoid mode, in which a relief valve controlled by a solenoid is opened and closed to regulate pressure. Both the pump and solenoid modes used a Proportional-Integral-Derivative (PID) control scheme. The system mode can also be in manual mode which allows the operator to manually control the pump and solenoid.

The next component is the communication network in which the protocol used is serial Modbus RTU. Modbus packets include a header and a payload. For Modbus over a Serial Line, a packet includes a device address, function code, payload, and a cyclic

redundancy code (CRC) or linear redundancy code (LRC). Modbus/TCP packets include a Modbus Application Protocol (MBAP) header, function code, and payload. The MBAP header includes a transaction identifier, protocol identifier, length, and device identifier. The device identifier is similar to the Modbus over Serial Line address. The data sets described in this work are taken from a Modbus over Serial Line; however, they can be safely used as a proxy for Modbus/TCP data with the exception there is no transaction identifier, protocol identifier, and length field. The diagram below provides a visual representation of a Modbus TCP and RTU packet.

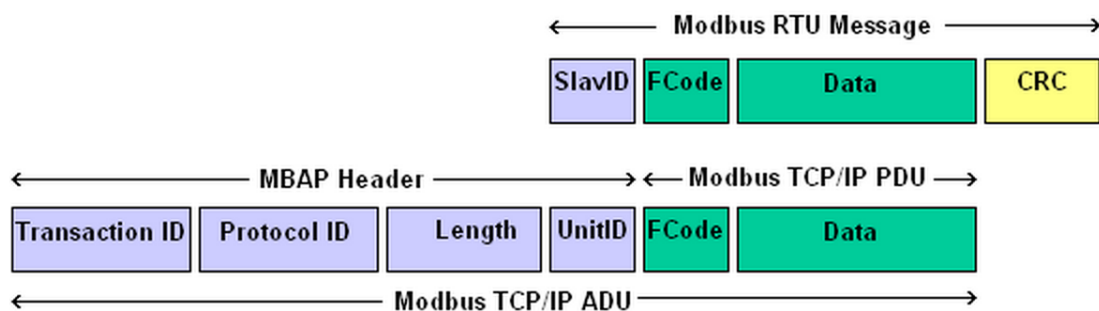


Figure 3.2 Modbus RTU and TCP packet [21]

The transaction identifier is generally a count of transaction numbers. The protocol identifier is always 0 for legal Modbus/TCP packets, and the length is the number of bytes in the payload plus 1 byte for the function code.

Inside, the payload Modbus/TCP and Modbus over Serial Line packets are identical. Modbus read and write commands are the most common command types. Read and write payloads include additional packet attributes such as coil or register addresses, quantities of requested or returned coils or registers, coil or register contents, error codes,

and exception codes. Some exceptional commands, such as the Diagnostic, file record access, mask write, and read FIFO commands include sub function codes, and other attributes to describe specific queries and responses.

The last component in the gas pipeline is the supervisory controls. These include the MTU and the iFIX HMI. The MTU is set up in a one-to-many configuration, meaning that all slave devices (RTUs) receive their controls from the one MTU, and the many RTUs respond back to the MTU. The HMI connects to the MTU and provides an interface for a human operator to monitor the system and provide supervisory controls when needed.

In the next sections of this chapter the process in which the dataset was collected, along with a detailed description of the dataset, are provided. The chapter will also include a discussion to show that unintentional trends have been eradicated from the dataset.

3.4 Dataset Collection Methodology

A new method of providing stimulus and collecting data logs was used to create the dataset. The first step for improving the dataset was to parameterize and randomize the order in which the attacks were executed. The execution was done by taking all attacks from [7] and implementing them in a man-in-the-middle fashion. The purpose of the man-in-the-middle method is to include all types of attacks as shown in the diagram below.

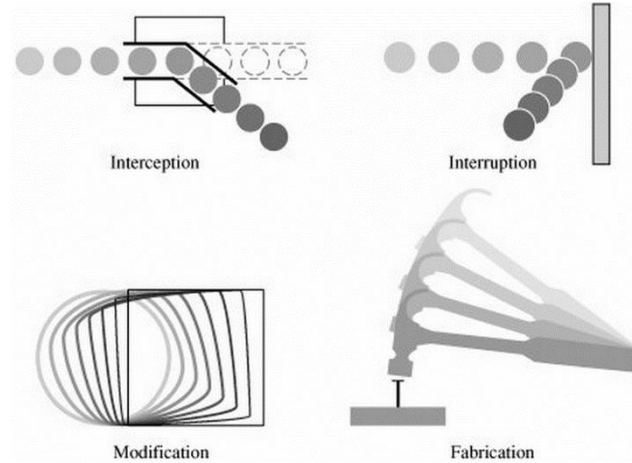


Figure 3.3 Types of Cyber Threats[27]

Interception refers to attacks which are sent both to the attacker and to the initial receiver. These types of attacks allow attackers to gain information about the each node's protocols, normal operation, and even the brand and model of the RTUs that the system is utilizing. Interruption attacks are used to block all communication between two nodes in a system. This type of attack would be a Denial of Service (DoS) between the MTU and the RTU slave device in the gas pipeline. The modification attacks allow an attacker to modify parameters or states in a system. In terms of the gas pipeline, an attacker could modify the set point parameter exclusively and leave all other parameters untouched. Similarly, the fabrication attacks allow for execution by completely creating a new packet to be sent between the MTU and RTU. The attacks in the gas pipeline dataset fit into these categories, but are broken down even further. The categories of attacks contained in the dataset are shown in the table below.

Table 3.1 Attack Categorization

Type of Attacks	Abbreviation	Threat Type
Normal	Normal(0)	N/A
Naïve Malicious Response Injection	NMRI(1)	Modification/Fabrication
Complex Malicious Response Injection	CMRI(2)	Modification/Fabrication
Malicious State Command Injection	MSCI(3)	Modification/Fabrication
Malicious Parameter Command Injection	MPCI(4)	Modification/Fabrication
Malicious Function Code Injection	MFCI(5)	Modification/Fabrication
Denial of Service	DoS(6)	Interruption
Reconnaissance	Recon(7)	Interception

The parameterization was accomplished by establishing ranges for which each attack operates. These ranges are created to provide a coverage of all possible attacks that could be executed on a specific parameter. For example, the set point manipulation attack modifies the set point parameter that controls the pressure level in the gas pipeline. This attack ranges from extreme levels to ranges that are within normal operations. Once the parameterization of each attack was accomplished, an algorithm to execute the attacks in a random order was designed.

The algorithm's intent is to allow for all attacks to execute an equal number of times and to reduce the unintended patterns that were found in the first iteration of the dataset. This does not necessarily mean the same number of attack packets will be created or modified for all attacks, as some attacks require fewer packets to execute while others require more to execute. For example, the function code scan attack is intended to scan all function codes which exist in the Modbus framework. The number of packets required for this attack will be significantly higher than the number to manipulate the set point

parameter. Once randomization of the attack patterns was implemented, the states that were considered normal are also randomized. To accomplish the randomization of normal states, an Auto IT script was coded to allow direct interaction with the iFIX HMI. Auto IT is a Windows scripting language which allows programmers to automate interaction with GUI's, in this case a HMI. It is able to simulate mouse movements and keyboard inputs of a process control network operation. The HMI controls and displays information regarding the gas pipeline. It provides a visual representation of the current state and operation of the gas pipeline. The Auto IT script simulates an operator changing the system state and the PID parameters. During testing of the system, there are physical constraints which prevent the pump from constantly being turned on. Therefore, the script must allow the pump to have a cool down time of twenty minutes and a running time of seven minutes. The script in turn runs the system at a 25.9% duty cycle.

A datalogger was also implemented to record only the packets which are received by either the MTU or the RTU. The data logger sits on the man-in-the-middle PC and was directly integrated into the attack framework through the use of C file input and output. Figure 3.3 below illustrates the entire process.

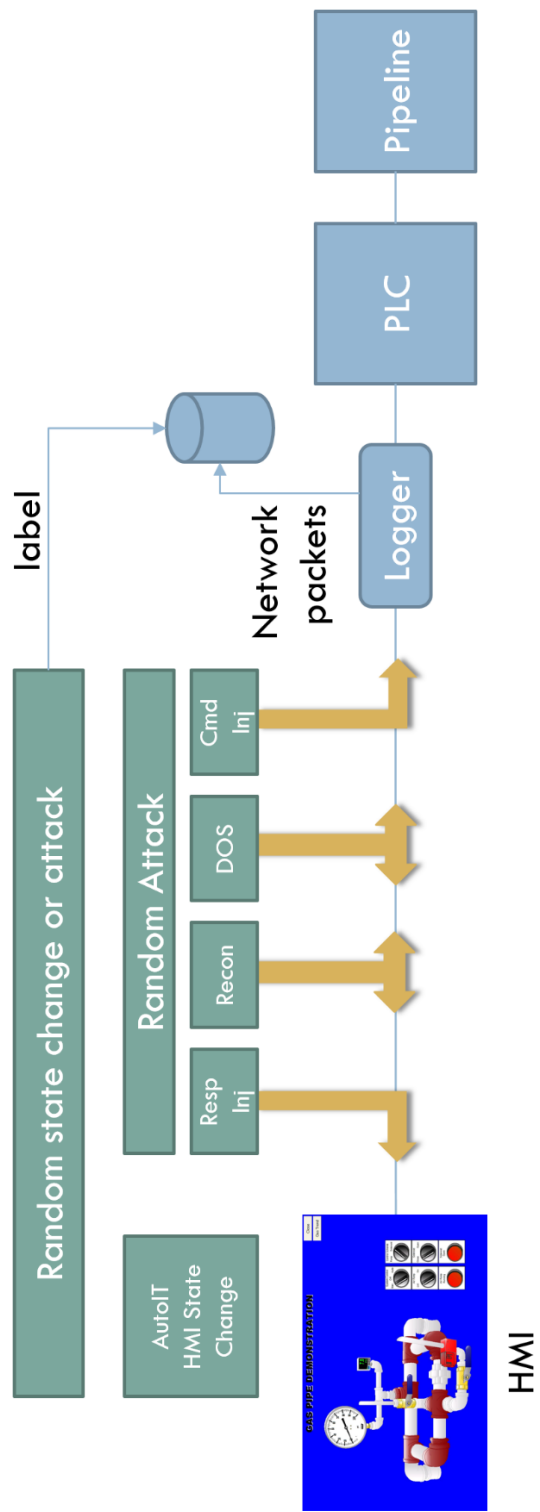


Figure 3.4 Process Framework/Methodology

3.5 Dataset Description

The datasets provided from this work are in two forms. The first form is a comma separated value (CSV) text file. The second form is an Attribute Relationship File Format (ARFF). The ARFF dataset was created to be used with WEKA. Waikato Environment for Knowledge Analysis, WEKA, is a tool which has a comprehensive list of machine learning algorithms [28]. WEKA has been used by many researchers in the IDS field for testing the performance of specific algorithms. The organization of the dataset provided represents one packet being delivered to either the MTU or to the RTU. Each instance in the dataset contains network traffic information along with payload information. The network information provides a pattern of communication for intrusion detection systems to train against. Unlike Information Technology (IT) networks, SCADA systems have network topologies which are fixed and the transactions between the nodes are repetitive and regular. This static behavior is conducive to IDSs to detect anomalous activity. The second category of features is that of payload information. The payload information provides information about the gas pipeline's state, settings, and parameters. These values are vital to understanding how the system is performing and detecting if the system is in an out-of-bounds or critical state. There are a total of 274,627 instances in each dataset. Each row in the dataset contains multiple columns, which are commonly referred to as features. These features will be discussed further in detail below. The consequence of representing each Modbus frame as a row in the dataset is that not all frames contain the same information, and many features are unknown for some instances.

3.5.1 Raw Dataset

A raw unprocessed dataset is provided. The dataset contains raw network traffic data. The purpose of providing the raw data is to provide a way of validating the legitimacy of the preprocessed, ARFF dataset, and to allow for researchers to preprocess with their own specialized methods. There are six features for each instance in the raw dataset. The first feature contains the Modbus frame that was received by either the master or slave device. The Modbus frame contains all information from the network, state, and parameters of the gas pipeline. The frame can be processed by determining the function code which the system is using and utilizing the memory mapping provided in Appendix A. The diagram in Appendix A contains the register values for both the master and slave side PLCs. The memory mapping also provides the information that is contained within each register such as set point, PID parameters, and state information. The frame can be preprocessed into separate features for each register on the PLC. The diagram below details an example Modbus frame for a write command from the MTU to the RTU.

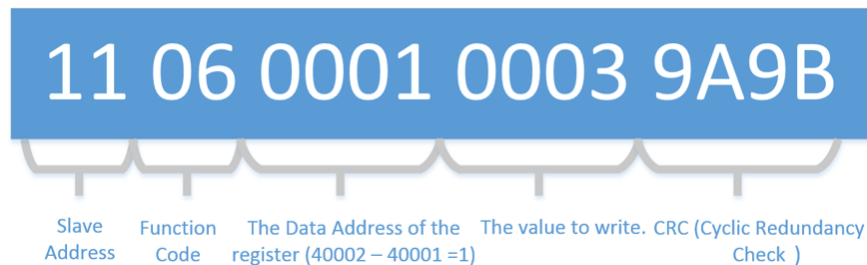


Figure 3.5 Modbus Frame

The register that is being written to in this write command is register 40002. In the Modbus protocol, read and write register values start at 40000. These registers contain the state and parameter information for the entire gas pipeline system. The ARFF dataset provides some features that have been extracted from these register locations.

The second and third feature in a raw dataset row represent the category of attack and specific attack that was executed. The specific category values are described in Tables 3.5, 3.6, 3.7, and 3.8. The second feature is the major category (Table 3.5) and the third feature is the specific attack (Tables 3.6, 3.7, and 3.8). In the case of a normal operation Modbus frame, both of these features will report a zero. Both of these features are necessary to train a supervised learning algorithm, as they allow the algorithm to learn the behavior of these attack patterns. The diagram later in the section will provide a one to one representation of the label and description to the categories and specific attacks.

The fourth and fifth features in a raw dataset row represent the source and destination of the frame. There are only three possible values for the source and destination feature. The value can be a '1,' which represents the master device sent the packet, a '2,' meaning the man-in-the-middle computer sent the packet, or a '3,' which means the slave device sent the packet. The purpose for this field is to provide a label to explain the origin of the packet and to aid in the preprocessing of the raw dataset. The last feature in the raw dataset contains a time stamp. The time stamp can be used to calculate a time interval which could be used to help with an IDS. In normal operation of the system, the time interval may only marginally change, but malicious command injection

or modification may lead to a larger time interval change. The figure below provides an example row from the raw dataset.

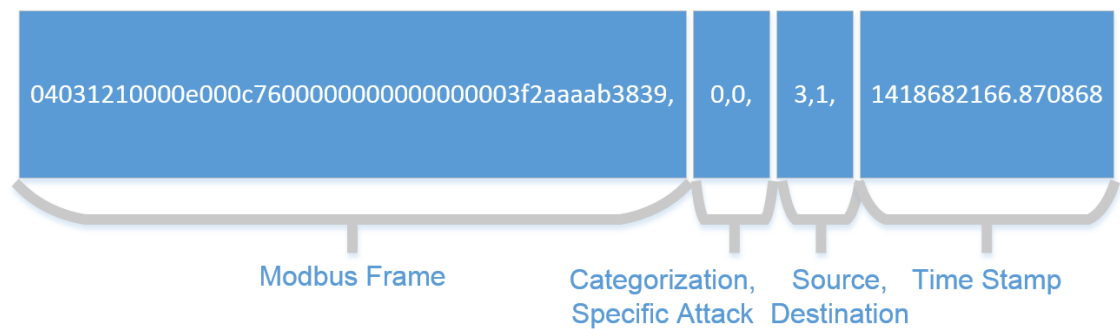


Figure 3.6 Instance within Raw Dataset

3.5.2 ARFF Dataset

The ARFF dataset was created to be used with WEKA. It contains twenty features, some of which are the same as in the raw dataset. The table below lists all twenty features.

Table 3.2 Feature List

Features	
address	control scheme
function	pump
length	solenoid
setpoint	pressure measurement
gain	crc rate
reset rate	command response
deadband	time
cycle time	binary result
rate	categorized result
system mode	specific result

The first feature contains the station address of the slave device. The station address is a unique eight bit value that is assigned to each master and slave device. The address is used to identify the slave that the master is transmitting commands to and the slave which is responding. The Modbus protocol is configured so that all slave devices receive all master transactions. The slave device must check the station address field to determine if the message is intended for itself or for a different slave device. This feature is used to enhance detection of device scan attacks, which broadcast commands to all possible station addresses to determine which addresses are operable. The second feature contains the function code. The function codes primarily used in the gas pipeline are read (0x03) and write commands (0x16), but a possibility of 256 different function codes exist. Some of these function codes can be used for malicious purposes, such as function code '0x08'. The '0x08' function code is generally used for diagnostics purposes, but it

can be used to force a slave device into a listen only mode. An attack like this would cause a denial of service using a valid function code. IDSs can utilize this feature to detect function codes which are out of the ordinary. The third feature contains the Modbus frame length. Similar to the function code, the length of the Modbus frame is fixed for each command or response query. In the gas pipeline system, a set of write and read commands are used to repeatedly perform block writes and block reads from specific registers. In detection of attacks, frames which are not of a specific length are easily detected as anomalous.

The fourth feature contains the set point value that controls the pressure in the gas pipeline. The set point feature is utilized when the gas pipeline system mode is set to 'automatic'. The slave ladder logic attempts to maintain the set point value provided by either opening a solenoid valve or turning on and off the pump. The set point feature effects the physical system drastically and would be a common point of malicious intent for an attacker. The next five features represent the PID controller values. Gain, reset rate, dead band, cycle time, and rate are all values which are used to tune the PID controller. An error is calculated based on these five parameters and allows the PID controller to open and close the relief valve or turn on and off the pump to minimize the error.

The tenth feature contains the value which controls the system's duty cycle. There are only three possible values which are valid and are shown in the table below.

Table 3.3 System mode features

System Mode Feature	
0	Off
1	Manual
2	Automatic

Since the gas pipeline is configured to have a 25.9% duty cycle, the system mode feature is generally set to '0' unless the system is active. The eleventh feature in the dataset is the control scheme feature. The control scheme in the gas pipeline determines whether the system will be controlled by the pump or by the solenoid. If the control scheme is set to pump, '0', the solenoid will remain opened and the pump is cycled to maintain gas pressure at the set point. The pump will continue to pump against the opened solenoid which simulates a load in a real gas pipeline. If the pump is set to solenoid, '1', the pump is constantly on and the pressure is controlled by opening and closing a solenoid valve to allow pressure to escape.

The twelfth feature controls the pump state only if system mode is set to manual. The feature can only be two values off, '0', or on, '1'. The system can be put into a critical state if an attacker were able to change the system mode to manual and turn the pump on. This type of attack could over pressurize the system and cause serious physical damage. The thirteenth feature controls the state of the solenoid valve when the system is also in manual mode. There are only two possibilities for this feature '0', closed, and '1', opened. Similar attacks to that of the pump feature could cause serious damage to the system by over or under pressurizing the system. The fourteenth feature contains the

current pressure measurement from the gas pipeline. The measurement is being provided by a pressure gauge attached to the pipeline and the data is stored in a register. The register is read by the master device and displayed on the HMI. This feature can be used in many attacks to provide a false measurement to imitate behavior that is not actually occurring in the system. The fifteenth feature contains the cyclic redundancy check (CRC). The cyclic redundancy check allows the system to check for errors within a frame that is being provided to either the master or the slave device. An attacker could constantly transmit a bad CRC to cause a DoS style attack. In Modbus-TCP, the CRC feature does not exist, the CRC is provided by the TCP frame. The sixteenth feature is provided to allow an IDS to learn the difference between commands and responses. The value can either be a '0' for response or '1' for command. This information is not parsed from the Modbus frame itself, but rather is provided to aid in the preprocessing step. The last four features, time stamp, specific attack, category attack, and binary attack, were also provided in the raw dataset. Tables 3.5 -3.6 detail all of the features with their respective type that are provided in the dataset.

Table 3.4 Feature list

Feature	Type
address	Network
function	Command Payload
length	Network
setpoint	Command Payload
gain	Command Payload
reset rate	Command Payload
deadband	Command Payload
cycle time	Command Payload
rate	Command Payload
system mode	Command Payload
control scheme	Command Payload
pump	Command Payload
solenoid	Command Payload
pressure measurement	Response Payload
crc rate	Network
command response	Network
time	Network
binary attack	Label
categorized attack	Label
specific attack	Label

As discussed in the introduction, Flame, Stuxnet, and Aurora have caused tremendous worry and have called in to question the security of current SCADA systems. Cyber threat and vulnerability research show the security challenges that SCADA

systems face and demonstrate many different approaches. As discussed in Chapter II researchers analyzed many different attack vectors and showed the security challenges that face the SCADA systems. In these papers, the researchers demonstrated several categories of attacks such as command injection, reconnaissance, and denial of service on current SCADA protocols. Since these protocols are open standards it allows everyone to study all angles of attack and provide security solutions. In order to provide a dataset for SCADA IDS research many of these types of attacks must be executed against the SCADA system. The attacks that are used in this data were found in Gao's [7] research. Gao developed seven categories of attacks. A table of these categories is provided below.

Table 3.5 Categories of attacks [7]

Type of Attacks	Abbreviation
Normal	Normal(0)
Naïve Malicious Response Injection	NMRI(1)
Complex Malicious Response Injection	CMRI(2)
Malicious State Command Injection	MSCI(3)
Malicious Parameter Command Injection	MPCI(4)
Malicious Function Code Injection	MFCI(5)
Denial of Service	DoS(6)
Reconnaissance	Recon(7)

The seven categories of attacks are split into four overall categories: command injection, response injection, denial of service (DoS), and reconnaissance. The description of the attacks was found in Gao's work and is summarized below [30]. All attacks have been slightly modified in this work, but are similar in behavior. The

command injection attacks contain malicious state command injection (MSCI), malicious parameter command injection (MPCI), and malicious function code injection attacks (MFCI). The response injection attacks provide two types of behaviors. The first is naïve malicious response injection (NMRI) which has sporadic and out of bounds behavior that would not be present in normal operation. These attacks generally occur when the malicious attacker lack information about the physical system process. The second type of response injection is complex malicious response injection (CMRI). These attacks leverage the state and physical process information to design attacks which mimic certain normal behaviors.

Table 3.6 Cyber attacks 1-12

Attack Name	Number	Type	Description
Setpoint Attacks	1-2	MPCI	Changes the pressure set point outside and inside of the range of normal operation.
PID Gain Attacks	3-4	MPCI	Changes the gain outside and inside of the range of normal operation.
PID Reset Rate Attacks	5-6	MPCI	Changes the reset rate outside and inside of the range of normal operation.
PID Rate Attacks	7-8	MPCI	Changes the rate outside and inside of the range of normal operation.
PID Deadband Attacks	9-10	MPCI	Changes the dead band outside and inside of the range of normal operation.
PID Cycle Time Attacks	11-12	MPCI	Changes the cycle time outside and inside of the range of normal operation.

The next category of attacks are reconnaissance attacks. Reconnaissance attacks are designed to collect information about the system through some passive gathering, or by forcing information from a device. The information can include network information (state address, length, crc, etc.), or device characteristics (model number, communication protocol, manufacturer, supported function codes).

CMRI attacks provide a level of sophistication over that of NMRI attacks. They mimic certain behaviors which occur within normal bounds. These injected states are leveraged to cause the system to lose efficiency, or cause loss of product and money. These attacks can be used to hide state changes which can occur in command injection attacks. Since these attacks inject states which display normal operation they become more difficult to detect.

Table 3.7 Cyber attacks 13-23

Attack Name	Number	Type	Description
Pump Attack	13	MSCI	Randomly changes the state of the pump.
Solenoid Attack	14	MSCI	Randomly changes the state of the solenoid.
System Mode Attack	15	MSCI	Randomly changes the system mode.
Critical Condition Attacks	16-17	MSCI	Places the system in a Critical Condition. This condition is not included in normal activity.
Bad CRC Attack	18	DoS	Sends Modbus packets with incorrect CRC values. This can cause denial of service.
Clean Registers Attack	19	MFCI	Cleans registers in the slave device.
Device Scan Attack	20	Recon	Scan for all possible devices controlled by the master.
Force Listen Attack	21	MFCI	Forces the slave to only listen.
Restart Attack	22	MFCI	Restart communication on the device.
Read Id Attack	23	Recon	Read ID of slave device. The data about the device is not recorded, but is performed as if it were being recorded.

MSCI, MPCl, and MFCI attacks inject control configuration commands to modify the system state and behavior. There are several impacts command injection attacks can cause such as “loss of process control, interruption of device communications, unauthorized modification of device configurations, and unauthorized modification of process set points” [30]. MSCI attacks are designed to modify the state of the current physical process. These types of attacks can place the system into a critical state which can cause harm to the system and even the lives of the operators. MPCl attacks modify

parameters which determine set point and PID configurations. MFCI attacks inject commands which exploit network protocol commands to change the behavior of the network. Denial of Service (DoS) attacks attempt to disrupt communications between the control and the process. This can be done through interruption of wireless networks, or network protocol exploits.

Table 3.8 Cyber attacks 24-35

Attack Name	Number	Type	Description
Function Code Scan Attack	24	Recon	Scans for possible functions that are being used on the system. The data about the device is not recorded, but is performed as if it were being recorded.
Rise/Fall Attacks	25-26	CMRI	Sends back pressure readings which create trends on the pressure reading's graph.
Slope Attacks	27-28	CMRI	Randomly increases/decreases pressure reading by a random slope
Random Value Attacks	29-31	NMRI	Random pressure measurements are sent to the master.
Negative Pressure Attack	32	NMRI	Sends back a negative pressure reading from the slave.
Fast Attacks	33-34	CMRI	Sends back a high set point then a low setpoint which changes "fast"
Slow Attack	35	CMRI	Sends back a high setpoint then a low setpoint which changes "slow"

Tables 3.6-3.8 provide a list and description for each of the 35 attacks within the dataset. Many of the specific types of attacks run on this system are general and can be applied to almost all types of systems. Therefore, this dataset can be used for research inside and outside of the industrial control system area. The next section will discuss why the proposed dataset is more suitable for IDS research than Gao's dataset.

3.6 Dataset Validation

This section provides a detailed validation of why the new dataset has improved from the previous dataset. The validation is provide by calculations comparing the two datasets.

The dataset was run through a subset of the tests that are found in [26] to determine if the patterns from Gao’s dataset have been eradicated. This paper was written by Zac Thornton, Jeff Hsu, and David Mudd to determine if machine learning algorithms could be used for anomaly detection in SCADA systems. They also wanted to determine how effective these machine learning algorithm by testing them with the Gao dataset. [26]. The end results showed that the Gao’s gas pipeline dataset contained unintended patterns. This same procedure was followed to determine if the new dataset contained similar patterns. Since the dataset contains 275,000 instances, the algorithms required a significant amount of time and memory to execute. This was also expressed in Thornton et al.’s paper and was addressed by using a 10% subset of Gao’s dataset. The process used to acquire the 10% from the 100% was not conveyed in the paper, as such the algorithms used were chosen to reduce time and memory constraints.

Table 3.9 List of Algorithms

Algorithms	Category
Naïve Bayesian Network	Bayes
PART	Rule-Based
Random Tree	Decision Tree
Multilayer Perceptron	Neural Network

Step one in comparison of the old and new datasets was to use the datasets with the machine learning algorithms listed in Table 3.9. Classification accuracy results from the algorithms were collected and compared to that of Thornton et al.'s results. Table 3.10 below illustrates the differences between the two datasets.

Table 3.10 Results of Algorithms

Algorithm	New Dataset Classification Accuracy	Gao's Dataset Classification Accuracy
Naïve Bayesian Network	80.39%	98.5%
PART	94.14%	99.32%
Random Tree	99.7%	99.9%
Multilayer Perceptron	85.22%	100%

Table 3.10 shows the algorithms have become less accurate at detecting anomalies using the new datasets and this is the direct result of the new methodology that was used to create the dataset. Classification accuracy cannot be the only statistic analyzed when determining the effectiveness of algorithms. False positive (FP) rates, precision, and recall are equally as important. False positive is an important statistic, as it can reveal discrepancies when it comes to the percentage of normal activity vs attack activity. For example, if system A has 99% of all traffic being normal while only 1% of the traffic being considered anomalous, then an IDS could consider all traffic to be normal and achieve a true positive rate of 99% which may sound good, but in reality it did not detect any of the 1% of anomalous traffic. The table below shows the percentage of attack and normal traffic in each dataset.

Table 3.11 Percentage of attacks in dataset

Dataset	Percentage of Attack Instances	Percentage of Normal Instances
New Dataset	21.9%	78.1%
Gao's Dataset	37.1%	62.8%

The discrepancy between normal and attack scenarios is shown through the kappa statistic. The kappa statistic provides a metric to quantitatively show the agreement between two observers. The statistic shows the percentage of agreements between two observers who randomly assign each instance in the dataset a label. The kappa statistic for the dataset is 83.1%. This means that 83.1% of all randomly assigned instances by the observers (knowing the percentage of attack vs normal traffic) match-up.

Further analysis was conducted using the PART algorithm. The PART algorithm was chosen as it is a rule-based algorithm which is well suited for a fixed-network topology that has regular communication patterns. It was used because it highlighted the many differences which show the benefits of the proposed dataset. It also provides results which detail exactly which categories of attacks have reduced patterns. Further analysis was also conducted using the three other algorithms, but is not provided in this research. The results from the other algorithms support the results of the PART algorithm. Tables 3.12-3.14 show exactly which categories of attacks were not detected using the PART algorithm.

Table 3.12 Comparison of False Positive Rates

Category	New Dataset FP (%)	Gao's Dataset FP (%)
Normal	20.7%	1.1%
NMRI	.8%	0%
CMRI	.5%	.1%
MSCI	0%	0%
MPCI	0%	.2%
MFCI	0%	0%
DoS	0%	0%
Recon	0%	0%

Table 3.12 shows that 20.7 % of attack traffic is falsely reported as normal and not as a category of attack compared to that of 1.1 % in the Gao dataset. This result shows an improvement from the Gao dataset because the attacks are harder to decipher from normal without further preprocessing or feature selection techniques. Inspection of precision and recall reveals the exact attack categories in the new dataset which are being classified incorrectly. Precision is the ratio of the number of instances classified correctly as a category of attack and the total number of instances classified as that category of attack. The equation below shows how precision was calculated for NMRI attacks.

$$Precision = \frac{\# \text{ of instances classified correctly as NMRI}}{\text{total \# of instances classified as NMRI}} \quad (3.1)$$

Precision will provide a metric to determine the instances classified into a category of attack against how many are actually of that category.

Recall is the ratio of the number of instances that are classified correctly as a category of attack and the total number of instance in that category of attack. In the case of NMRI attacks, the calculated recall is shown below.

$$Recall = \frac{\# \text{ of instances classified correctly as NMRI}}{\text{total \# of NMRI instances}} \quad (3.2)$$

Recall provides a metric to determine the true positive ratio in a category of attack. The table below provides the values for recall and precision for both datasets.

Table 3.13 Precision and Recall for Datasets

	New Dataset		Gao's Dataset	
Category	Precision	Recall	Precision	Recall
Normal	94.5%	99.9%	99.4%	99.5%
NMRI	74.2%	82.4%	99.5%	94.4%
CMRI	89.3%	82.1%	99.4%	99.9%
MSCI	99.3%	54.9%	97.4%	95.1%
MPCI	99.8%	63.9%	97.5%	98.0%
MFCI	98.6%	100.0%	100.0%	95.8%
DoS	99.6%	48.3%	99.8%	97.9%
Recon	100.0%	97.1%	100.0%	100.0%

Table 3.13 shows that the precision and recall for all attack categories in the Gao dataset are high. Thus, there were very few instances where the PART algorithm was unable to provide the correct classification and only 1.1% of the attack instances were reported as normal conditions shown in Table 3.12. Table 3.13 also shows that the precision in the new dataset is the lowest for the response injection attacks. The low precision lies in the PART algorithms fault in differentiating between NMRI and CMRI

attacks. The confusion matrix below shows the misclassification of NMRI and CMRI attacks.

Table 3.14 Confusion Matrix for NMRI and CMRI attacks

Category	Predicted NMRI	Predicted CMRI
Actual NMRI	6389	1148
Actual CMRI	2156	10703

The confusion matrix provides a look into exactly how the PART algorithm categorized the two categories of attacks. The misclassification is due in part to the randomness of NMRI attacks, which has the possibility of overlapping in values with the CMRI attacks.

Table 3.12 also shows that the recall rates for DoS, MPCI, and MSCI are around 50%, but have high precision. This means that the instances considered to be these categories of attacks were indeed from these categories, but suffered in discovering all instances which were from each category of attack. In the case of the DoS attacks, only 48.3% of instances were classified correctly. The reason for the low recall was because of the Bad CRC attack. The Bad CRC attack injected an incorrect CRC value in a write multiple register function command, which would cause the RTU to ignore the command and in turn cause a DoS. The PART algorithm was able to correctly classify the write command with the incorrect CRC value as an attack, but failed to classify the response from the RTU as an attack. Therefore, the misclassification led to some skewing of the results, and in reality, the detection of these attacks is greater than 98%. The approximation was calculated by taking the precision into account and realizing that

99.6% of the commands that were thought to be DoS attacks were actually categorized as DoS attacks. These result shows that the precision and recall for both the Gao dataset and the new dataset are similar for the DoS attacks. The PART algorithm was still able to detect the behavior of DoS attacks successfully and the new attack framework did not change this fact, which is expected.

The low recall rates found in the MPCl and MSCl categories of attacks are a direct result of the new attack framework. Since the system is now placed into all possible normal conditions, the algorithms are forced to differentiate between multiple normal conditions instead of just three that were provided in the old dataset. In the case of the MPCl attacks, the set point and PID parameter coverage is greater in the new dataset. Figure 3.7 highlights the differences in how the coverage of set point values is different from the previous iteration of the dataset.

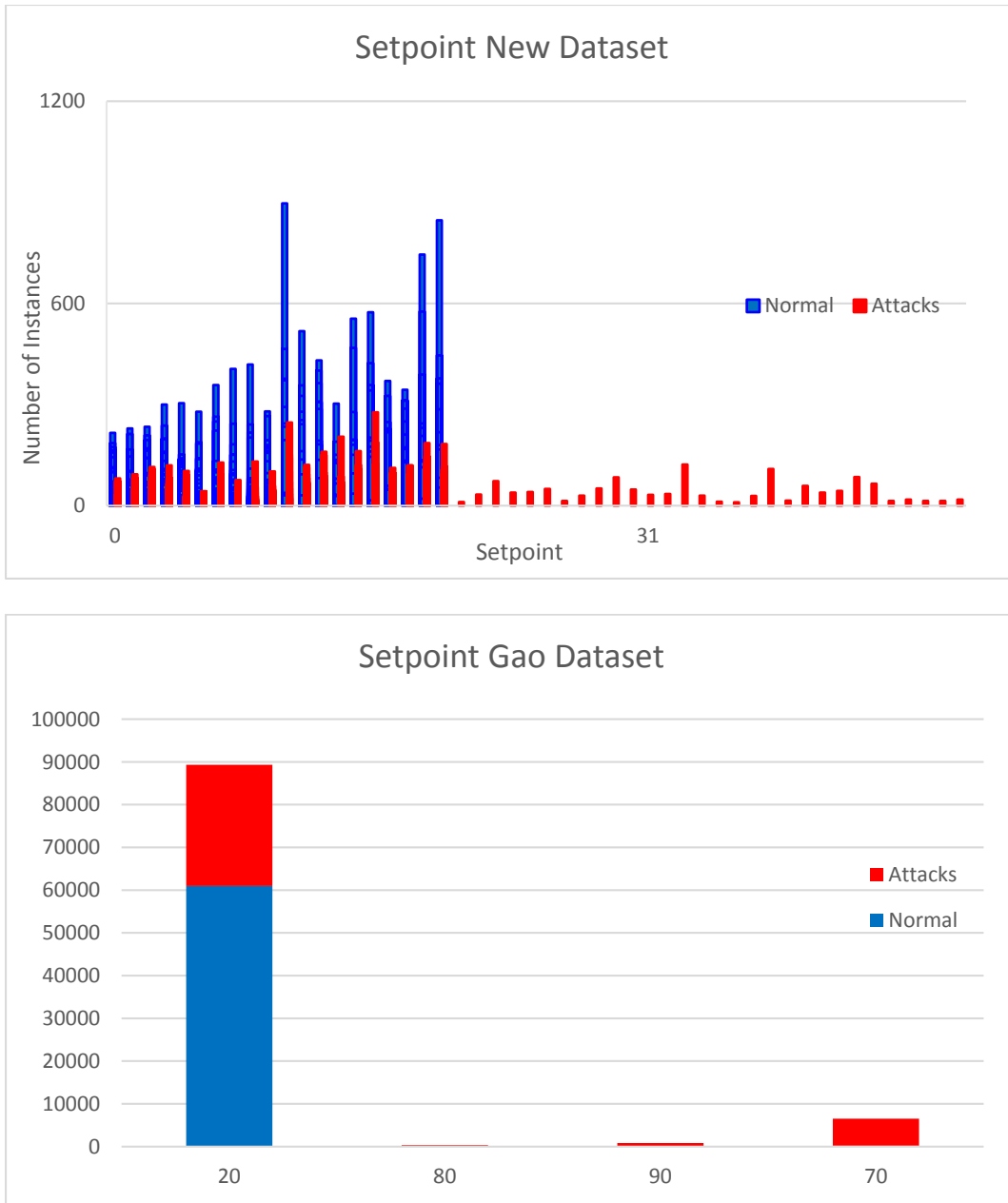


Figure 3.7 Differences between Setpoint Values

The bars labeled blue are normal behavior and the bars labeled red represent a packet that contains an attack. The previous iteration of the dataset only allowed the system to be in one normal configuration which was at 20 PSI. If the RTU did not receive

20 PSI as the set point value, then it was blatantly obvious that the system was being attacked. Thornton et al. expressed this in the future work section of their report, “Setpoint only has unique 4 values: 20, 70, 80, and 90” [26]. Since the attacks were easily detected, the high detection rates seen in Table 3.10 are unrealistic. This static behavior is not only found in this feature of the dataset, but in most of the PID parameters as well. Figure 3.8 illustrates the behavior in the gain PID parameter.

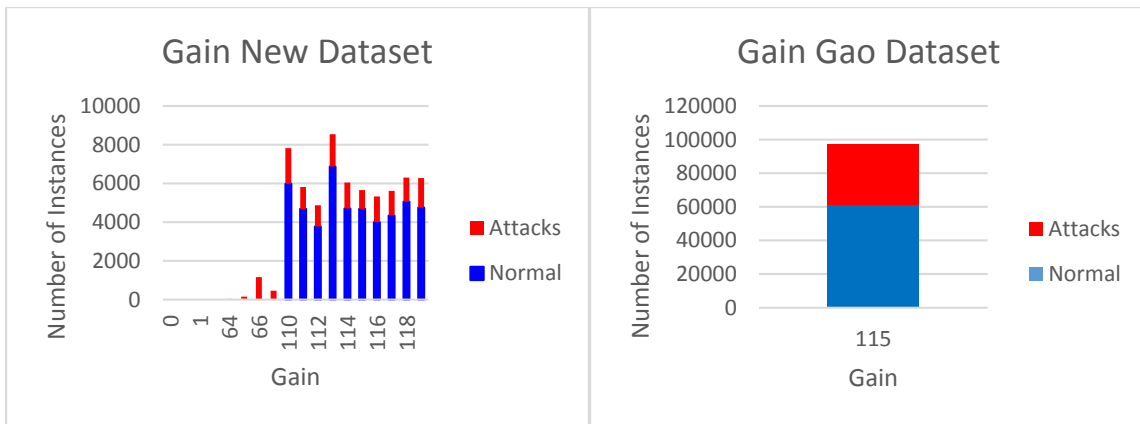


Figure 3.8 Gain Parameter Coverage

The new iteration provides more coverage by providing a range of values for each parameter and is the direct cause to the lower detection rate.

The MSCI attacks were also effected in a similar way. The system was not being placed into all system control modes and schemes which was causing easily detectable patterns in the Gao dataset. This problem was mentioned in [26]. Figure 3.9 shows how the system is now placed into all control modes.

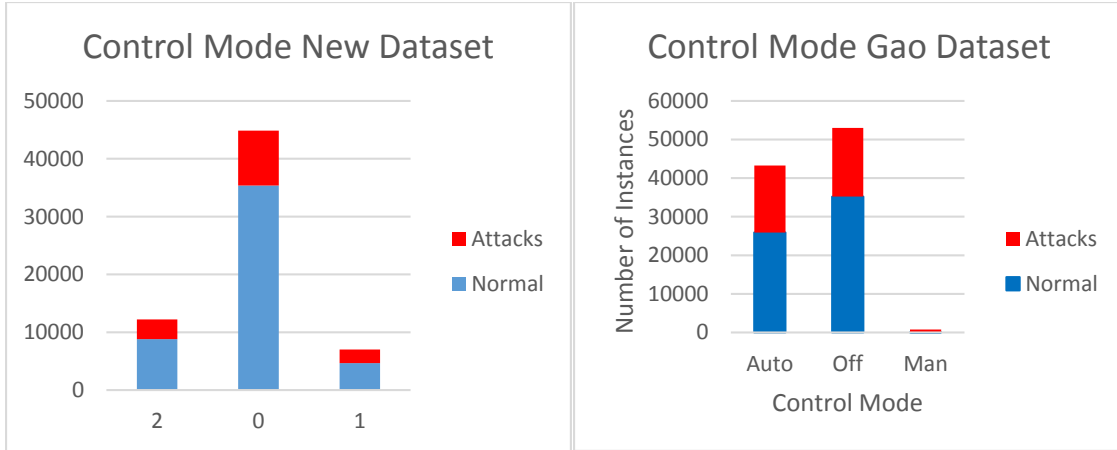


Figure 3.9 Differences between Control Modes

Now that the system is placed into all system modes, the measurement values are more reflective of real behavior and have more variance. This also limits the number of obvious attacks that were prevalent in the previous iteration of the dataset. Table 3.15 is from Thornton et al.'s work and describes all the problems that were found in the previous dataset.

Table 3.15 Identified Problems in the Feature list of Gao's Dataset [26]

command_address	setpoint	control_mode
Always 4, unless DoS attack	always 20 unless MPCFI attack	only 1 when MSCFI
reponse_address	resp_read_fun	control scheme
always 19 unless Recon attack	only 1 when normal or CMRI attack	only 0 when MSCFI
comm_read_function	subfunction	measurement
always 3 unless DoS attack	always 0 unless MFCFI attack	All CMRIs in range 6-11 All NMRIs grossly out of bounds

Table 3.15 describes many of the obvious correlations between the features and the attack patterns. The setpoint, res_read_fun, control_mode, control scheme, subfunction, and measurement have been addressed by using the framework that was described in the previous section and have shown to reduce the ease of detection. Some of correlations are inherent to the system because of the repetitiveness of its interaction with the slave device. These correlations should be embraced by the machine learning algorithms to aid in detecting the anomalous behavior.

CHAPTER IV

CONCLUSIONS

SCADA systems are becoming more vulnerable to outsider threats with increased network connectivity. The need for industrial control system IDS research is increasing. This thesis provides a set of labeled network data logs captured while a laboratory scale gas pipeline was in normal states and under cyber-attacks as described. A new methodology for implementing the attacks and a simulated operator have been implemented to create these data logs. The data logs include artifacts of 35 cyber-attacks and can be used to train and test classifiers used by IDSs. Comparison to a previous iteration of the dataset was conducted to provide validation that the dataset has improved. This was done by showing that obvious correlations between features and attack patterns have been removed. The datasets will facilitate comparison of different IDS implementations and provide third-party validation of results.

REFERENCES

- [1] M. Pires; P.S. Oliveira, L.A.H.G., "Security Aspects of SCADA and Corporate Network Interconnection: An Overview," Dependability of Computer Systems, 2006. DepCos-RELCOMEX '06. International Conference on , vol., no., pp.127,134, 25-27 May 2006 doi: 10.1109/DEPCOS-RELCOMEX.2006.46
- [2] S. Cherry, How Stuxnet Is Rewriting the Cyberterrorism Playbook. 2010. Available at: <http://spectrum.ieee.org/podcast/telecom/security/how-stuxnet-is-rewriting-the-cyberterrorism-playbook> accessed on 09.05.2014.
- [3] W. Jones, "Flame: Cyberwarfare's Latest, Greatest Weapon." - IEEE Spectrum. IEEE, May 2012. Web. 27 Oct. 2014.
- [4] J. Meserve, "Sources: Staged Cyber Attack Reveals Vulnerability in Power Grid." CNN. Cable News Network, Sept. 2007. Web. 27 Oct. 2014.
- [5] J. Weiss, "Misconceptions about Aurora: Why Isn't More Being Done." InfoSec Island. N.p., 13 Apr. 2012. Web. 27 Oct. 2014.
- [6] "Introduction to Industrial Control Networks" (PDF). IEEE Communications Surveys and Tutorials. 2012.
- [7] T. Morris; W. Gao, Industrial Control System Network Traffic Data Sets to Facilitate Intrusion Detection System Research, Mississippi State University.
- [8] M. Brundle; M. Naedele "Security for process control systems: An overview", IEEE Security Privacy, vol. 6, no. 6, pp.24 -29 2008
- [9] H. Sugwon; L., Myongho; , "Challenges and Direction toward Secure Communication in the SCADA System," Communication Networks and Services Research Conference (CNSR), 2010 Eighth Annual , vol., no., pp.381-386, 11-14 May 2010
- [10] D. Kang; J. Lee; S. Kim; J. Park; "Analysis on cyber threats to SCADA systems," Transmission & Distribution Conference & Exposition: Asia and Pacific, 2009 , vol., no., pp.1-4, 26-30 Oct. 2009
- [11] "Dell Security Annual Threat Report." Boom: A Journal of California 5.1 (2015): 12-13. Dell. Dell, 2015. Web. 5 May 2015.

- [12] S. Valentine and C. Farkas "Software security: Application-level vulnerabilities in SCADA systems," Information Reuse and Integration (IRI), 2011 IEEE International Conference on , vol., no., pp.498-499, 3-5 Aug. 2011
- [13] D. Dzung; M. Naedele.; V. Hoff, T.P.; Crevatin, M.; "Security for Industrial Communication Systems," Proceedings of the IEEE , vol.93, no.6, pp.1152-1177, June 2005
- [14] "Understanding Intrusion Detection." Sans.org. SANS Institute, 2001. Web. 27 Oct. 2014.
- [15] M. Pires, P.S.; Oliveira, L.A.H.G., "Security Aspects of SCADA and Corporate Network Interconnection: An Overview," Dependability of Computer Systems, 2006. DepCos-RELCOMEX '06. International Conference on , vol., no., pp.127,134, 25-27 May 2006 doi: 10.1109/DEPCOS-RELCOMEX.2006.46
- [16] R. Anderson (2001). Security Engineering: A Guide to Building Dependable Distributed Systems. New York: John Wiley & Sons. pp. 660-667.
- [17] A. Almalawi, X. Yu, Z. Tari, A. Fahad, I. Khalil, "An unsupervised anomaly-based detection approach for integrity attacks on SCADA systems", Computers & Security, Volume 46, October 2014, Pages 94-110, ISSN 0167-4048,
- [18] J.M. Moya; Á. Araujo; Z. Banković; J.-M.De Goyeneche.; J.C. Vallejo; Malagón, P.; D. Villanueva; D. Fraga; E. Romero; J. Blesa, Improving Security for SCADA Sensor Networks with Reputation Systems and Self-Organizing Maps. Sensors 2009, 9, 9380-9397.
- [19] J. Carr. "Snort: Open Source Network Intrusion Prevention." ESecurity Planet. ESecurity Planet, 5 June 2007. Web. 02 Nov. 2014.
- [20] H. Lin , A. Slagell , C. Di Martino , Z. Kalbarczyk , R.K. Iyer, Adapting Bro into SCADA: building a specification-based intrusion detection system for the DNP3 protocol, Proceedings of the Eighth Annual Cyber Security and Information Intelligence Research Workshop, January 08-10, 2013, Oak Ridge, TN
- [21] "Simply Modbus - About Modbus TCP." Simply Modbus - About Modbus TCP. N.p., n.d. Web. 03 June 2015. <<http://www.simplymodbus.ca/TCP.htm>>.
- [22] A. Mahmood; H. Jianku; Z. Tari,;Y. Xinghuo; , "Building a SCADA Security Testbed," Network and System Security, 2009. NSS '09. Third International Conference on , vol., no., pp.357-364, 19-21 Oct. 2009
- [23] S. Cheung et al. "Using model-based intrusion detection for SCADA networks." Proceedings of the SCADA security scientific symposium. Vol. 46. 2007.

- [24] D. Yang, A. Usynin, and J. Wesley Hines. "Anomaly-based intrusion detection for SCADA systems." 5th intl. topical meeting on nuclear plant instrumentation, control and human machine interface technologies (npic&hmit 05). 2006.
- [25] K. Da 2000. Attack development for intrusion detection. Master's Thesis. Massachusetts Institute of Technology, Cambridge, MA.
- [26] Z. Thornton, J. Hsu, and D. Mudd. SCADA ANOMALY DETECTION. Rep. Mississippi State University, n.d. Web.
<http://www.ece.msstate.edu/~morris/icsdatasets/MSU_SCADA_Final_Report.pdf>.
- [27] "Cryptography and Security in Computing." (2012): n. pag. Tech Target. Web.
- [28] M. Hall, E. Frank, G. Holmes, B. Pfahringer, P. Reutemann, I. H. Witten (2009); The WEKA Data Mining Software: An Update; SIGKDD Explorations, Volume 11, Issue 1.
- [29] Boyer, Stuart. "Collecting Data from Distant Facilities." ISA. International Society of Automation, 27 Oct. 2014. Web. Oct. 2007.
- [30] Gao, W., Morris, T., Reaves, B., Richey, D. On SCADA Control System Command and Response Injection and Intrusion Detection, in the Proceedings of 2010 IEEE eCrime Researchers Summit. Dallas, TX. Oct 18-20, 2010. Best Paper Award Winner!

APPENDIX A

REGISTER MAPPING FOR GAS PIPELINE RTU

MASTER REGISTER	SLAVE REGISTER	SLAVE NAME	DESCRIPTION	DATA TYPE	RANGE
READS					
43011	43000	Gas	Digital Outputs	Binary	0,1
43012	43001	Gas	Digital Inputs	Binary	0,1
43013	43002	Gas	Analog Input 0	Integer	0-32767
43014	43003	Gas	Analog Input 1	Integer	0-32767
43015	43004	Gas	Analog Input 2	Integer	0-32767
43016	43005	Gas	Analog Input 3	Integer	0-32767
43017	43006	Gas	Analog Input 4	Integer	0-32767
43018	43007	Gas	Scaled Gas Pressure	Float	0-100.0 psi
43019	43008	Gas			
43516	43050	Gas	HMI Control Scheme Status	Integer	0=Pump, 1=Solenoid
43517	43051	Gas	HMI System Control Status	Integer	0=Off, 1=Man, 2=Auto
43518	43052	Gas	HMI Manual Pump Run Command	Integer	0=Off, 1=Run
43519	43053	Gas	HMI Manual Solenoid Open Command	Integer	0=Close, 1=Open
43520	43054	Gas	Spare		
43521	43055	Gas	Spare		
43522	43056	Gas	PID Setpoint	Float	0-50 psi
43523	43057	Gas			
43524	43058	Gas	PID Gain	Float	0-100
43525	43059	Gas			
43526	43060	Gas	PID Reset	Float	0-100
43527	43061	Gas			
43528	43062	Gas	PID Rate	Float	0-100
43529	43063	Gas			
43530	43064	Gas	PID Deadband	Float	0-100
43531	43065	Gas			
43532	43066	Gas	PID Cycle Time	Float	0-100
43533	43067	Gas			

Figure A.1 Register Mapping Sheet